

The Annotated Rudin

Principles of Mathematical Analysis, Third Edition

Walter Rudin

COMPANION

The Implication Lattice

Rudin's text appears here complete and unaltered — his sentences, his paragraph breaks, his numbering. Nothing has been paraphrased away. The annotation is strictly additive: numbered markers in his prose key to notes in the outer margin, and longer commentary sits in ruled boxes between his paragraphs, where it can be skipped without breaking his argument.

WHAT THIS DOCUMENT IS

FIVE WORDS, AND THE ARROWS BETWEEN THEM

Has a limit, continuous, differentiable, integrable, convergent.

Every course in analysis uses all five, and the relations among them are the real content of Rudin's Chapters 3–7 — but they are never assembled in one place, because each is introduced where it is needed and the connections are made in passing.

This document assembles them. For every ordered pair of properties it records a verdict: either an implication, with a proof, or a non-implication, with the function that kills it. Nothing is left ambiguous, and nothing is asserted without one or the other.

A structural point that shapes everything. Four of the five properties are *local* — they are properties of a function at a point. Riemann integrability is not; it is a property of a function on a whole interval, and asking whether f is integrable *at* p is meaningless. A single five-by-five table would quietly equivocate between the two. So the material splits into three blocks:

Block A — the local properties at a point p .

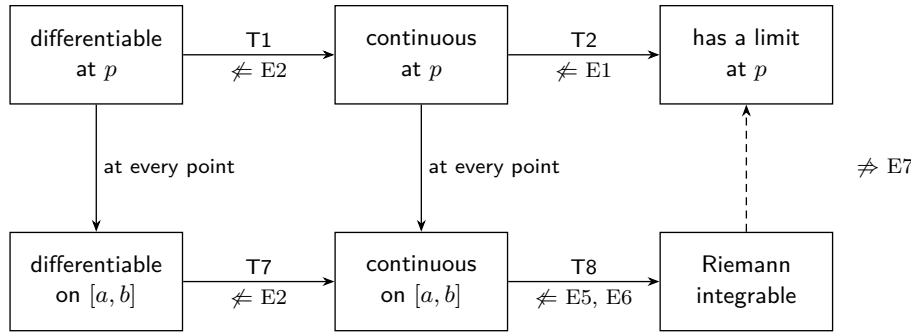
Block B — those properties promoted to “at every point of $[a, b]$,” together with integrability.

Block C — what survives a limit: given $f_n \rightarrow f$, which properties of the f_n are inherited by f ?

The spine. Block C is not a new subject. It is Blocks A and B asked a second time with a limit in front, and the answers change — which is precisely why uniform convergence had to be invented.

Numbering. Implications are **T1–T18** and are proved where they are stated. Counterexamples are **E1–E12**; each has a dossier in the last section. References of the form *S2.13* or *Exercise 4.18* point into *The Annotated Rudin*; plain numbers like *4.19* are Rudin's own.

THE LATTICE, AT A GLANCE



Blocks A and B. Solid arrows are theorems, proved below; the label beneath each records the failure of its converse and names the function that witnesses it. Bounded is not enough for integrability (E7), and integrability implies nothing pointwise — an integrable function may be discontinuous on a dense set (E6).

READ THE DIAGRAM AS THREE CLAIMS | One: the top row is a strict chain, and each strictness is cheap — one explicit function apiece. **Two:** the bottom row is *not* the top row repeated. Integrability is genuinely global, and the arrow into it comes from compactness, not from anything pointwise. **Three:** there is no arrow back from integrability. Integrable functions can be badly discontinuous, and bounded functions can fail to be integrable altogether.

BLOCK A — AT A POINT

WHAT IS BEING COMPARED | Fix p , a limit point of the domain of f . Three properties are in play: $\lim_{x \rightarrow p} f(x)$ exists; f is continuous at p ; f is differentiable at p . They form a strict chain, and the two strictnesses cost one function each.

T1 Theorem. *If f is differentiable at p then f is continuous at p .*

Proof. For $x \neq p$,

$$f(x) - f(p) = \frac{f(x) - f(p)}{x - p} \cdot (x - p).$$

As $x \rightarrow p$ the first factor tends to $f'(p)$ and the second to 0, so the product tends to 0 by the algebra of limits (4.4). Hence $f(x) \rightarrow f(p)$.

□

This is Rudin's 5.2, and the proof is worth one remark: it uses the existence of $f'(p)$ only to know the quotient has *some* finite limit.¹

T2 Theorem. *If f is continuous at p then $\lim_{x \rightarrow p} f(x)$ exists and equals $f(p)$.*

Proof. Definition 4.5 says precisely this whenever p is a limit point of the domain. □

¹ So the theorem is really “if the difference quotient is bounded near p then f is continuous at p ” — differentiability is more than is needed. That observation is what makes the Lipschitz condition of Chapter 4's Supplement E the natural weakening.

At an *isolated* point of the domain, f is continuous automatically (4.6) while $\lim_{x \rightarrow p} f(x)$ is not defined at all — Rudin’s 4.1 requires p to be a limit point. So T2 needs the hypothesis, and the arrow in the diagram is an arrow only over limit points.

This is the deleted-versus-undeleted distinction of Chapter 4’s Supplement A (S4.2). Rudin’s limit is deleted, which is exactly why continuity is not literally the statement “the limit equals the value” — at isolated points there is no limit to equal it.

Both converses fail. **E1**, the removable spike, has a limit at 0 and is not continuous there; **E2**, $|x|$, is continuous at 0 and not differentiable there.

T3 Theorem. (*Darboux.*) *If f is differentiable on $[a, b]$ then f' takes every value between $f'(a)$ and $f'(b)$ — even though f' need not be continuous.*

Proof. Suppose $f'(a) < \lambda < f'(b)$ and put $g(x) = f(x) - \lambda x$, so $g'(a) < 0 < g'(b)$. Since $g'(a) < 0$, $g(t) < g(a)$ for some t just right of a ; since $g'(b) > 0$, $g(s) < g(b)$ for some s just left of b . So the minimum of the continuous g on the compact $[a, b]$ (4.16) is attained at an interior point x , and there $g'(x) = 0$ by 5.8 — that is, $f'(x) = \lambda$. $\square$

T4 Corollary. *A derivative has no simple discontinuities. In particular the signum function is not the derivative of anything.*

Proof. A simple discontinuity is one at which both one-sided limits exist (Definition 4.26). If they existed and differed, f' would omit the values strictly between them on either side of the point, contradicting T3; if they existed and agreed but differed from $f'(x)$, the same argument applies to a value between $f'(x)$ and the common one-sided limit. $\square$

What T3 and T4 buy, read against Chapter 4’s taxonomy

Chapter 4’s appendix opened Rudin’s two kinds of discontinuity into four (S4.19): removable, jump, asymptotic, oscillatory. T4 says a derivative may suffer only the last two. It cannot jump, and it cannot have a removable discontinuity.

E3, the function $x^2 \sin(1/x)$, shows the permitted case really occurs: it is differentiable everywhere, and its derivative oscillates without limit at 0. So “differentiable” and “continuously differentiable” are genuinely different hypotheses — which is why Rudin states them separately from Chapter 5 onward, and why $\mathcal{C}^1$ needs its own name.

T5 Theorem. *If $\lim_{x \rightarrow p} f(x)$ exists then f is bounded on some punctured neighbourhood of p .*

Proof. Take $\varepsilon = 1$ in Definition 4.1: there is $\delta > 0$ with $|f(x) - q| < 1$, hence $|f(x)| < |q| + 1$, for all x in the punctured δ -neighbourhood. $\square$

Boundedness is the weakest thing in this section, and it is strictly weaker

than everything above it: **E4**, $\sin(1/x)$, is bounded everywhere and has no limit at 0.²

BLOCK B — ON AN INTERVAL

WHAT CHANGES WHEN THE POINT IS DROPPED | Promote each local property to “at every point of $[a, b]$ ” and a fourth property joins them: Riemann integrability, which has no pointwise form at all. The arrow into it does not come from anything local — it comes from compactness.

2 The chain therefore extends one step downward: *differentiable* $\Rightarrow$ *continuous* $\Rightarrow$ *has a limit* $\Rightarrow$ *locally bounded*, with all three implications strict. Rudin never states T5, but he uses it — at 4.4, where the product of limits needs one factor bounded, and again at 5.2.

T7 Theorem. *If f is differentiable at every point of $[a, b]$ then f is continuous on $[a, b]$.*

Proof. T1 at each point. □

T8 Theorem. *If f is continuous on $[a, b]$ then f is Riemann integrable there.*

Proof. Let $\varepsilon > 0$. Since $[a, b]$ is compact, f is *uniformly* continuous (4.19), so there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon/(b-a)$ whenever $|x - y| < \delta$. Take any partition P of mesh less than δ . On each subinterval $[x_{i-1}, x_i]$ the supremum M_i and infimum m_i are attained (4.16), at points less than δ apart, so $M_i - m_i \leq \varepsilon/(b-a)$ and

$$U(P, f) - L(P, f) = \sum_i (M_i - m_i) \Delta x_i \leq \frac{\varepsilon}{b-a} \sum_i \Delta x_i = \varepsilon.$$

By Rudin’s criterion 6.6, f is integrable. □

HOW THE PROOF FLOWS

- 1. Choose the method.** Integrability is a statement about upper and lower sums being close, so the target is $U - L < \varepsilon$. That is a statement about the *oscillation* of f on the pieces of a partition, so the whole problem is: make the oscillation uniformly small.
- 2. Spend compactness — once.** Plain continuity gives a δ at each point, and those δ ’s could shrink to nothing as the point moves. Uniform continuity is exactly the statement that one δ serves everywhere, and 4.19 is where compactness is spent. Everything after this is bookkeeping.
- 3. Convert to the partition.** Mesh smaller than δ makes every subinterval a place where the uniform estimate applies.
- 4. Sum.** The Δx_i add to $b - a$, which is why the $\varepsilon/(b-a)$ was chosen in step 2 — reverse-engineered from the final line, the same way Rudin picks his constants at 1.21.
- 5. Audit.** Compactness is spent once, in step 2; the attainment of

M_i and m_i (4.16) is a convenience, not a necessity — $\sup - \inf \leq \varepsilon/(b-a)$ would follow from the uniform estimate anyway. Note what is *not* used: nothing about differentiability, and nothing pointwise beyond continuity.

WHICH FUNCTIONS ARE INTEGRABLE

FROM A GALLERY TO A THEOREM | The examples below could each be settled by hand with upper and lower sums. Doing so leaves the impression that integrability is a matter of luck. It is not: there is an exact criterion, and it is stated in the language Chapter 2's and Chapter 4's appendices already built.

T9 Definition. A set $E \subset \mathbb{R}$ has **measure zero** if for every $\varepsilon > 0$ there are countably many intervals $I_1, I_2, \dots$ with $E \subset \bigcup_k I_k$ and $\sum_k |I_k| < \varepsilon$.

countably many intervals, total length $< \varepsilon$



the set E — may be infinite, even dense

Measure zero. The intervals may be as numerous as you like, so long as their lengths add to less than ε — and ε is arbitrary. Every countable set qualifies: cover its k th point by an interval of length $\varepsilon/2^{k+1}$. No interval $[a, b]$ with $a < b$ qualifies.

T10 Theorem. (Lebesgue's criterion.) A bounded f on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has measure zero.

Proof sketch. Recall from S2.5 and S4.5 that with $\Omega_\varepsilon = \{x : \omega_f(x) \geq \varepsilon\}$, each Ω_ε is closed — hence compact here — and the discontinuity set is $D = \bigcup_n \Omega_{1/n}$.

($\Rightarrow$) Let f be integrable and fix n . Given $\eta > 0$ choose a partition with $U - L < \eta/n$. Every subinterval whose interior meets $\Omega_{1/n}$ contributes at least $(1/n)\Delta x_i$ to $U - L$, so those subintervals have total length less than η . They cover $\Omega_{1/n}$ apart from finitely many partition points, which are covered by intervals of arbitrarily small total length. So each $\Omega_{1/n}$ has measure zero, and D , a countable union of such sets, does too.

($\Leftarrow$) Suppose D has measure zero and $|f| \leq M$. Fix $\varepsilon > 0$. Then Ω_ε is compact and of measure zero, so finitely many open intervals of total length $< \varepsilon$ cover it. Off their union, every point has $\omega_f < \varepsilon$, hence a neighbourhood on which the oscillation is below ε ; compactness of the remaining closed set gives a finite subcover and so a partition on which those pieces each oscillate by less than ε . Splitting $U - L$ into the two kinds of subinterval bounds it by $2M\varepsilon + \varepsilon(b-a)$, which is arbitrarily small. $\square$

The criterion settles the whole row at once

Everything in the integrability row now follows by inspection of a discontinuity set:

function	discontinuous on	integrable?
step function (E5)	a finite set	yes
Thomae (E6)	$\mathbb{Q}$ — countable	yes
Dirichlet (E7)	everything	no
$x^2 \sin(1/x)$ (E3)	nowhere	yes

Two things are worth dwelling on. First, E6 is integrable while being discontinuous on a *dense* set — so “integrable” is compatible with very bad pointwise behaviour, and there is no arrow at all from integrability back to continuity. Second, the machinery this proof runs on is not new: the closedness of Ω_ε and the decomposition $D = \bigcup_n \Omega_{1/n}$ are S2.5 and S4.5, proved in the appendices for quite different reasons. Oscillation was the right idea twice.

T11 Theorem. *Bounded does not imply integrable, and integrable does not imply continuous anywhere in particular.*

Proof. E7 is bounded and discontinuous everywhere, so by T10 it is not integrable. E6 is integrable by T10 and is discontinuous at every rational. $\square$

THE DEEPEST CELL: A DERIVATIVE THAT IS NOT INTEGRABLE

T12 Theorem. (Volterra.) *There is a function F on $[0, 1]$, differentiable at every point, whose derivative F' is bounded but **not** Riemann integrable.*

The construction, and what it costs the Fundamental Theorem

The construction starts from a *fat Cantor set* K — built like Rudin’s 2.44 but removing middle intervals of rapidly shrinking length, so that K is nowhere dense yet does *not* have measure zero. On each interval removed one plants a scaled copy of $x^2 \sin(1/x)$ (E3), tapered so that it and its derivative vanish at the endpoints, and sets $F = 0$ on K .

The resulting F is differentiable everywhere, with $|F'| \leq 1$. But at every point of K the derivative oscillates: arbitrarily close by there are points where F' takes values near ± 1 . So $\Omega_1 \supset K$, and K does not have measure zero — by T10, F' is not integrable.

What this costs. Rudin’s 6.21 says: if f is integrable on $[a, b]$ and $F' = f$, then $\int_a^b f = F(b) - F(a)$. Volterra’s example shows the hypothesis “ f is integrable” cannot be dropped — there really are derivatives to which the formula does not apply, because the integral on the left does not exist. So the two halves of the Fundamental

Theorem are independent statements: having an antiderivative does not make a function integrable, and being integrable does not make a function a derivative (E5, a step function, is integrable and is not a derivative, by T4).

FOUR PROPERTIES THAT SOUND ALIKE

Keep these apart. For a bounded f on $[a, b]$:

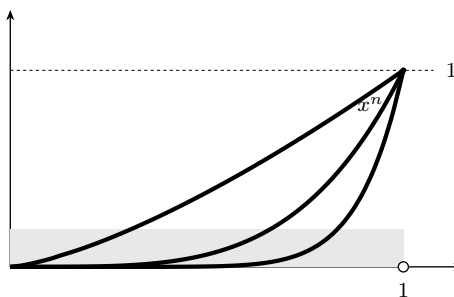
- (a) f is continuous;
- (b) f is Riemann integrable;
- (c) f has an antiderivative;
- (d) f is a derivative *and* integrable, so that 6.21 applies.

(a) implies all the others. (b) and (c) are independent: E5 is integrable with no antiderivative (T4), and Volterra's F' has an antiderivative and is not integrable. (d) is their conjunction, and it is what the Fundamental Theorem actually requires.

BLOCK C — WHAT SURVIVES A LIMIT

THE SAME QUESTIONS, WITH A LIMIT IN FRONT | Let $f_n \rightarrow f$. Each property of Blocks A and B can now be asked twice: do the f_n have it, and does f ? Pointwise convergence answers “no” almost every time. Uniform convergence — Rudin's 7.7 — was invented precisely to change the answers, and it changes two of the three.

T13 Definition. $f_n \rightarrow f$ *pointwise* on E if $f_n(x) \rightarrow f(x)$ for each $x \in E$; *uniformly* if $\sup_{x \in E} |f_n(x) - f(x)| \rightarrow 0$ — one N serving every x at once.



shaded: the ε -tube about the limit function

$f_n(x) = x^n$ on $[0, 1]$ (E9). The pointwise limit is 0 on $[0, 1)$ and 1 at 1 — discontinuous. However large n is, the graph still climbs out of any ε -tube about the limit before reaching 1, so the convergence is not uniform. Continuity is lost exactly because the N that works depends on x .

T14 Theorem. (Rudin 7.12.) If each f_n is continuous and $f_n \rightarrow f$ uniformly, then f is continuous.

Proof. Fix x_0 and $\varepsilon > 0$. Choose n with $\sup |f_n - f| < \varepsilon/3$, then δ with $|f_n(x) - f_n(x_0)| < \varepsilon/3$ for $|x - x_0| < \delta$. For such x ,

$$|f(x) - f(x_0)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(x_0)| + |f_n(x_0) - f(x_0)| < \varepsilon.$$

□

The three terms are where the name “ 3ε argument” comes from.³

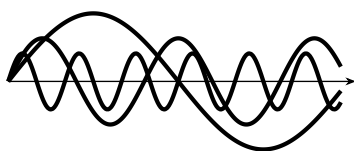
T15 Theorem. (Rudin 7.16.) If each f_n is integrable on $[a, b]$ and $f_n \rightarrow f$ uniformly, then f is integrable and

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

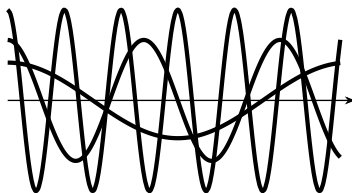
Proof. Put $\varepsilon_n = \sup_{[a,b]} |f_n - f| \rightarrow 0$. Then $f_n - \varepsilon_n \leq f \leq f_n + \varepsilon_n$, so for any partition $U(P, f) - L(P, f) \leq U(P, f_n) - L(P, f_n) + 2\varepsilon_n(b - a)$. Choosing n with $2\varepsilon_n(b - a)$ small and then P making the first term small gives integrability by 6.6. The same inequalities bound $|\int f - \int f_n|$ by $\varepsilon_n(b - a) \rightarrow 0$. □

T16 Theorem. Uniform convergence does **not** preserve differentiability, and even when the limit is differentiable, $\lim f'_n$ need not be f' .

Proof. Take $f_n(x) = \sin(nx)/\sqrt{n}$ on $\mathbb{R}$ (E10). Then $\sup |f_n| = 1/\sqrt{n} \rightarrow 0$, so $f_n \rightarrow 0$ uniformly and the limit is as differentiable as anything could be. But $f'_n(x) = \sqrt{n} \cos(nx)$, which is unbounded and has no limit at any point — at $x = 0$, $f'_n(0) = \sqrt{n} \rightarrow \infty$ while $f' \equiv 0$. □



$$f_n = \sin(nx)/\sqrt{n} \rightarrow 0$$



$$f'_n = \sqrt{n} \cos(nx) \text{ diverges}$$

Why uniform convergence cannot control derivatives. On the left the amplitude collapses like $1/\sqrt{n}$; on the right the same functions differentiated, where the extra factor n beats it and the amplitude grows like $\sqrt{n}$. Uniform smallness of a function says nothing about its slope — a wiggle can be short and steep at the same time.

T17 Theorem. (Rudin 7.17 — the repair.) Suppose each f_n is differentiable on $[a, b]$, the sequence $(f_n(x_0))$ converges for some $x_0 \in [a, b]$, and (f'_n) converges uniformly. Then (f_n) converges uniformly to some f , and $f' = \lim_n f'_n$.

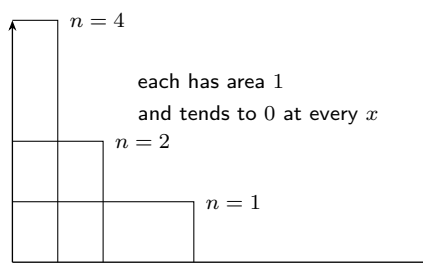
The hypothesis has moved: it is the *derivatives* that must converge uniformly, and the functions need only converge at a single point.⁴

T18 Theorem. Pointwise convergence preserves none of the three: not continuity, not integrability, and not the value of the integral.

3 Notice exactly what uniformity buys: the first and third terms. Under pointwise convergence one can still make each of them small, but only after fixing the point — and the middle term then needs a δ depending on the n that was chosen for that point, so the estimate never closes. E9 is what that failure looks like.

4 Read T16 and T17 together as one lesson. Differentiation is not continuous with respect to uniform convergence — so if you want to differentiate a limit, you must assume something about the derivatives, not about the functions. The same asymmetry returns for series at 7.17's corollary, and it is why term-by-term differentiation of a power series (8.1) needs its own theorem while term-by-term integration does not.

Proof. Continuity fails by E9. For integrability, enumerate $\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}$ and let f_n be the indicator of $\{q_1, \dots, q_n\}$; each f_n is integrable, having finitely many discontinuities (T10), and $f_n \rightarrow \text{Dirichlet}$ (E7) pointwise, which is not integrable. For the integral, take the spikes of E11: $f_n \rightarrow 0$ pointwise while $\int_0^1 f_n = 1$ for every n . $\square$



The spikes $f_n = n \cdot \mathbf{1}_{(0, 1/n)}$ (E11). At every fixed $x > 0$ the spike eventually moves left of x , so $f_n(x) \rightarrow 0$; at $x = 0$ the value is 0 throughout. Yet $\int f_n = 1$ always. The mass does not disappear — it escapes by getting narrower and taller, which no pointwise statement can see.

Two remarks the appendices already prepared

Pointwise limits are not arbitrary. It is tempting to conclude that pointwise convergence preserves nothing whatever. It preserves something: a pointwise limit of continuous functions is of Baire class 1, and not every function is. Dirichlet's function is *not* a pointwise limit of continuous functions — proved in Chapter 2's appendix at S2.13, using Baire's theorem. So E7 arises as a pointwise limit of the *integrable* f_n above, but never of continuous ones.

And uniform convergence is not the only repair. T15 requires uniformity, but the conclusion $\int \lim = \lim \int$ survives under far weaker hypotheses once the Lebesgue integral is available — dominated convergence needs only a pointwise limit and an integrable bound, which the spikes of E11 conspicuously lack. Rudin reaches that at 11.32. The uniformity in T15 is a limitation of the Riemann integral, not of the statement.

THE CAST

TWELVE FUNCTIONS CARRY THE WHOLE DOCUMENT

Each dossier gives the definition, the property the function has, the property it lacks, and the arrow it kills. Several are Rudin's own and are annotated in the main edition; those references are given.

	function	kills the claim that...
E1	removable spike	limit $\Rightarrow$ continuous
E2	$ x $	continuous $\Rightarrow$ differentiable
E3	$x^2 \sin(1/x)$	differentiable $\Rightarrow \mathcal{C}^1$
E4	$\sin(1/x)$	bounded $\Rightarrow$ has a limit
E5	step function	integrable $\Rightarrow$ continuous
E6	Thomae	integrable $\Rightarrow$ disc. set nowhere dense
E7	Dirichlet	bounded $\Rightarrow$ integrable
E8	Weierstrass	continuous $\Rightarrow$ differentiable anywhere
E9	x^n on $[0, 1]$	pointwise preserves continuity
E10	$\sin(nx)/\sqrt{n}$	uniform preserves differentiability
E11	tall spikes	pointwise preserves $\int$
E12	Volterra	derivative $\Rightarrow$ integrable

E1 Removable spike. $f(x) = 0$ for $x \neq 0$, $f(0) = 1$. Has limit 0 at 0; not continuous there, since the limit is not the value. The cheapest possible separation, and the reason Rudin's limit is *deleted* (S4.2).

E2 Absolute value. $f(x) = |x|$. Continuous everywhere; at 0 the difference quotient is $|h|/h = \pm 1$ according to the side, so the one-sided derivatives are -1 and $+1$ and $f'(0)$ does not exist. Note that f is Lipschitz, so even that stronger condition does not give differentiability.

E3 $x^2 \sin(1/x)$. $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. Then $|f(x)| \leq x^2$ forces $f'(0) = 0$, and for $x \neq 0$

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x},$$

which has no limit as $x \rightarrow 0$ — the second term oscillates between -1 and 1 . So f is differentiable everywhere with f' discontinuous at 0; the discontinuity is of the second kind, as T4 requires. This function is also the building block of E12.

E4 $\sin(1/x)$. Bounded by 1; no limit at 0, since $x_n = 1/(2n\pi)$ and $y_n = 1/(2n\pi + \pi/2)$ both tend to 0 while $f(x_n) = 0$ and $f(y_n) = 1$. Rudin's 4.27(d); the oscillatory case of the four-way taxonomy at S4.19, and the reason $\omega_f(0) = 2$.

E5 Step function. $f = 0$ on $[0, 1)$, $f = 1$ on $[1, 2]$. Integrable — one discontinuity, a set of measure zero (T10) — and not continuous. By T4 it is also *not* a derivative, since it has a jump. So E5 separates “integrable” from “has an antiderivative” in one direction, and E12 separates them in the other.

E6 Thomae. $f(p/q) = 1/q$ in lowest terms, $f = 0$ at irrationals. Continuous at every irrational and discontinuous at every rational. Inte-

grable by T10, because $\mathbb{Q}$ is countable and therefore of measure zero — with $\int_0^1 f = 0$. Rudin’s Exercise 4.18; S4.5 in the Chapter 4 appendix proves the companion fact that no function is continuous exactly on $\mathbb{Q}$, so Thomae has no mirror image.

E7 Dirichlet. $\mathbf{1}_{\mathbb{Q}}$. Bounded, discontinuous at every point, so $\omega_f \equiv 1$ and by T10 it is not integrable — directly, every upper sum is 1 and every lower sum is 0. It is the pointwise limit of the integrable indicators of finite rational sets (T18) but, by Baire, *not* of any sequence of continuous functions (S2.13).

E8 Weierstrass. $f(x) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi x)$ with $0 < a < 1$, b an odd integer, $ab > 1 + \frac{3}{2}\pi$. The series converges uniformly by the M -test, so f is continuous (T14); it is differentiable at no point. Rudin’s 7.18. Baire’s theorem (S2.13) upgrades the example to a statement about the typical case: the continuous functions differentiable somewhere form a set of first category in $\mathcal{C}[a, b]$, so nowhere differentiability is the rule and E2 the exception.

E9 x^n on $[0, 1]$. Each f_n is continuous; the pointwise limit is 0 on $[0, 1)$ and 1 at 1. Convergence is not uniform, since $\sup_{[0, 1]} |f_n - f| = 1$ for every n — take x just below 1. Kills “pointwise preserves continuity” (T18), and is the standard picture of a limit escaping an ε -tube.

E10 $\sin(nx)/\sqrt{n}$. $\sup |f_n| = 1/\sqrt{n} \rightarrow 0$, so $f_n \rightarrow 0$ uniformly; but $f'_n(x) = \sqrt{n} \cos(nx)$ diverges everywhere. Kills “uniform preserves differentiability” (T16). The mechanism is worth naming: differentiating multiplies the amplitude by the frequency, and uniform convergence bounds only the amplitude.

E11 Tall spikes. $f_n = n \cdot \mathbf{1}_{(0, 1/n)}$. Each is integrable with $\int_0^1 f_n = 1$; $f_n \rightarrow 0$ pointwise. Kills “pointwise gives $\int \lim = \lim \int$ ” (T18). The functions are not uniformly bounded, which is exactly the hypothesis the dominated convergence theorem (11.32) would need.

E12 Volterra. Built on a fat Cantor set K — nowhere dense, but not of measure zero — by planting a tapered copy of E3 in each removed interval and setting the function to 0 on K . The result is differentiable everywhere with bounded derivative, and the derivative oscillates at every point of K , so $\Omega_1 \supset K$ and by T10 the derivative is not integrable (T12). The single most informative example in the document: it shows the two halves of the Fundamental Theorem are independent, and it is why 6.21 must hypothesise integrability rather than deduce it.

WHAT TO CARRY AWAY

One. The local chain is strict at every link, and the witnesses are cheap — E1, E2, E3. Nothing subtle is needed to separate

limit, continuity, differentiability, and continuous differentiability.

Two. Integrability is not part of that chain. It is global, it comes from compactness (T8), and it has an exact characterisation in terms of the size of the discontinuity set (T10) — for which the oscillation machinery of S2.5 and S4.5 is precisely the right language.

Three. The interesting failures are all failures of *exchange*: $\lim$ against $\int$, $\lim$ against $\lim$, $\lim$ against $\frac{d}{dx}$. Uniform convergence fixes the first two and cannot fix the third, because differentiation amplifies exactly what uniformity controls.

Four. Where an implication fails, ask what hypothesis would restore it. Pointwise $\rightarrow$ uniform restores continuity and the integral; uniform on the functions $\rightarrow$ uniform on the derivatives restores differentiation (T17); Riemann $\rightarrow$ Lebesgue restores the integral exchange under a domination hypothesis instead of a uniformity one. Every gap in the lattice is an invitation of this kind.